

# AI-Assisted Threat Analysis Log

*Segment 2 — traffic\_segment\_2.csv*

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## 1. AI Tool Used

**Tool:** ChatGPT

ChatGPT was used as an AI-assisted security analysis tool to review the PCAP-derived CSV dataset and surface potentially suspicious patterns — communication relationships, protocol usage, and unusual traffic behaviour worth a closer look.

## 2. Dataset Information

**Dataset analyzed:** traffic\_segment\_2.csv

**Source:** Extracted from 2021-08-16-formbook.pcap

**Records analyzed:** 1,200 packets

The analysis set out to identify:

- Frequently communicating IP addresses
- Internal and external communication relationships
- Repeated communication patterns
- Potential automated behaviour
- Port and protocol usage
- Large data transfers
- Potential security concerns

## 3. AI Prompt Used

The following prompt was submitted to ChatGPT to begin the traffic investigation:

```
You are assisting with network traffic analysis as a cybersecurity analyst.
```

```
I have provided a CSV file extracted from a PCAP network capture.  
This is Segment 2 of a larger network traffic investigation.  
Analyze this dataset and identify potential security concerns.
```

```
Perform traffic investigation focusing on:
```

1. Identify the most frequently communicating source and destination IP addresses.
2. Determine internal/private and external/public IP addresses using standard IP classification methods.
3. Detect repeated communication patterns that may indicate automated activity, periodic communication, or beaconing behaviour.
4. Identify unusual external IP addresses or unexpected communication

relationships.

5. Analyze commonly used and unusual ports and protocols.
6. Identify large data transfers or repeated data exchange patterns.
7. Highlight traffic behaviour that requires further security investigation.

For every finding:

- Explain why the observation is relevant from a security perspective.
- Provide evidence from the dataset.
- Separate confirmed observations from assumptions.
- Do not label traffic as malicious without sufficient evidence.
- Suggest validation steps a SOC analyst should perform.

At the end provide:

1. Overall security assessment.
2. Confirmed observations.
3. Potential concerns with confidence level.
4. Recommended next investigation steps.

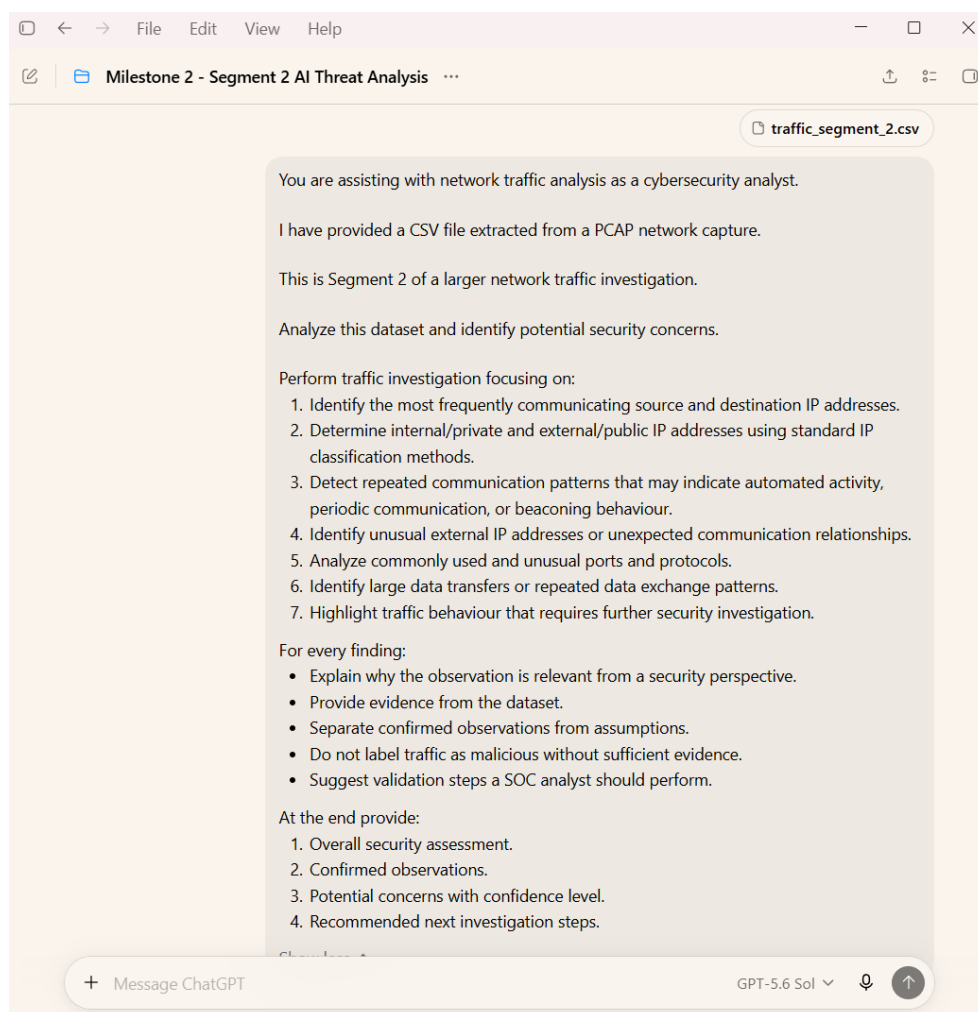


Figure 1: AI Prompt Submitted for Segment 2 Analysis

## 4. AI Generated Initial Findings (Segment 2)

### Segment 2 Network Traffic Investigation Report

**Dataset analyzed:** traffic\_segment\_2.csv

**Records analyzed:** 1,200 network frames

**Analysis scope:** IP communication patterns, address classification, ports/protocols, data exchange behaviour, and potential security concerns

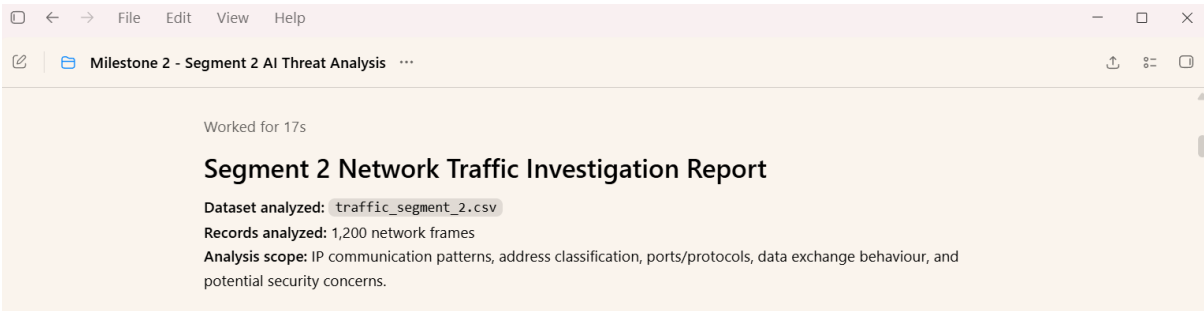


Figure 2: AI Generated Segment 2 Analysis Response

### 4.1 Most Frequently Communicating Source and Destination IP Addresses

#### Top Source IP Addresses

| Rank | Source IP      | Packets | Classification   |
|------|----------------|---------|------------------|
| 1    | 10.10.0.33     | 610     | Private/Internal |
| 2    | 18.184.26.60   | 543     | Public/External  |
| 3    | 204.79.197.200 | 42      | Public/External  |
| 4    | 93.184.220.29  | 3       | Public/External  |
| 5    | 8.8.8.8        | 1       | Public/External  |
| 6    | 13.107.42.23   | 1       | Public/External  |

10.10.0.33 is clearly the primary communicating system in the dataset, and its largest external relationship is with 18.184.26.60.

#### Top Destination IP Addresses

| Rank | Destination IP | Packets | Classification   |
|------|----------------|---------|------------------|
| 1    | 10.10.0.33     | 590     | Private/Internal |
| 2    | 18.184.26.60   | 575     | Public/External  |
| 3    | 204.79.197.200 | 31      | Public/External  |
| 4    | 93.184.220.29  | 2       | Public/External  |

|   |             |   |                   |
|---|-------------|---|-------------------|
| 5 | 8.8.8.8     | 1 | Public/External   |
| 6 | 10.10.0.255 | 1 | Private/Broadcast |

## 4.2 Internal and External IP Address Classification

Address classification used the standard RFC 1918 private ranges: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

### Internal Addresses Identified

| IP Address  | Classification            | Evidence                |
|-------------|---------------------------|-------------------------|
| 10.10.0.33  | Private host              | RFC1918 private range   |
| 10.10.0.255 | Private broadcast address | Local network broadcast |

### External/Public Addresses Identified

| IP Address     | Classification     | Evidence                        |
|----------------|--------------------|---------------------------------|
| 18.184.26.60   | Public Internet IP | Not part of private ranges      |
| 204.79.197.200 | Public Internet IP | Not part of private ranges      |
| 93.184.220.29  | Public Internet IP | Not part of private ranges      |
| 8.8.8.8        | Public Internet IP | Google DNS                      |
| 13.107.42.23   | Public Internet IP | Microsoft-related address range |

## 4.3 Repeated Communication Patterns and Possible Beaconing Behaviour

### Heavy repeated communication between internal host and external host

```

10.10.0.33 <---> 18.184.26.60
  Total packets:      1,118
  Internal -> External: 575 packets
  External -> Internal: 543 packets

Primary flow: 10.10.0.33:53480 -> 18.184.26.60:80
  Packets:      575
  Time range: 01:59:07 - 01:59:12

```

A connection this concentrated between one internal host and one external destination could mean any number of things — ordinary web browsing, a software update, an application checking in, or a download — but it's also the shape a command-and-control channel, data collection, or script-generated traffic can take. There isn't enough evidence here to call it malicious; what makes it worth

a second look is simply that one external IP accounts for almost all the traffic, the exchange is highly repetitive, and it all happens inside a short window.

### Beaconing assessment

Classic beaconing usually shows regular time intervals, small repeated packets, long-duration communication, and periodic callbacks. What's actually here looks different: most packets arrive within milliseconds of each other, the whole exchange spans about five seconds, and the timing is bursty rather than periodic — the average interval between packets is roughly 8 milliseconds. That doesn't line up well with classic malware beaconing; confidence for beaconing here is low.

## 4.4 Unusual External IP Addresses and Communication Relationships

| External IP    | Packets | Main Port   | Observation                     |
|----------------|---------|-------------|---------------------------------|
| 18.184.26.60   | 1118    | TCP/80      | Dominant external communication |
| 204.79.197.200 | 73      | TCP/443     | HTTPS communication             |
| 93.184.220.29  | 5       | TCP/80      | Small web communication         |
| 8.8.8.8        | 2       | UDP/53      | DNS query                       |
| 13.107.42.23   | 1       | Unknown TCP | Minimal communication           |

### External IP 18.184.26.60

```
10.10.0.33 -> 18.184.26.60
  Protocol:      TCP
  Destination port: 80
  Outbound packets: 575
  Inbound packets: 543
```

A single external server handling most of an internal system's traffic is worth investigating on its own — it represents a significant external dependency, and if it turns out to be unauthorized, the impact of a compromise could be substantial. A SOC analyst should confirm who owns the IP through WHOIS, ASN ownership, and threat intelligence reputation checks, and review the HTTP request content itself — User-Agent strings, URLs accessed, Host headers, and response data.

## 4.5 Ports and Protocol Analysis

### Protocol Distribution

| Protocol | Count |
|----------|-------|
| TCP      | 1171  |
| TCP/TLS  | 21    |
| UDP DNS  | 2     |

|            |   |
|------------|---|
| UDP NBNS   | 1 |
| HTTP media | 1 |

### TCP Port 80 (HTTP)

Port 80 accounts for 577 packets, almost all of it the 10.10.0.33:53480 → 18.184.26.60:80 flow. HTTP doesn't encrypt its payload, so credentials or other sensitive data could be exposed if any were transmitted, and malware commonly relies on HTTP for its own communication — though using HTTP by itself is not evidence of anything malicious.

### TCP Port 443 (HTTPS)

31 packets went to 204.79.197.200 over port 443, which looks like ordinary encrypted web traffic.

### UDP Port 53 (DNS)

A single DNS query went to 8.8.8.8, Google's public resolver — nothing here suggests DNS tunneling.

### UDP Port 137 (NBNS)

One NBNS packet went to the internal broadcast address 10.10.0.255, consistent with routine Windows network discovery. Low concern, since it stays internal.

## 4.6 Large Data Transfers and Data Exchange Patterns

**Total frame size:** 896,293 bytes (approximately 875 KB)

| Metric              | Value      |
|---------------------|------------|
| Minimum packet size | 54 bytes   |
| Average packet size | 746 bytes  |
| Maximum packet size | 1514 bytes |

Most of the traffic sits at the maximum Ethernet frame size of 1514 bytes, which points to bulk TCP exchange — web content, a file transfer, or something similar.

The 10.10.0.33 ↔ 18.184.26.60 flow accounts for 1,118 packets and roughly 800 KB of data. That could be a legitimate website download, application communication, or a software update, but it's also the kind of volume that warrants checking for unauthorized upload/download activity or exfiltration. Nothing in the dataset currently points to data theft.

## 4.7 Traffic Requiring Further Investigation

### Concern 1 — Dominant External Communication

18.184.26.60 accounts for 1,118 of the 1,200 captured packets — about 93% of all traffic. A compromised endpoint often shows this kind of repeated communication with a small number of external servers, which is why this pattern gets a medium confidence rating.

## Concern 2 — Unencrypted HTTP Communication

The 577 packets of HTTP traffic on port 80 travel in the clear, meaning any URLs, credentials, or application data inside could be exposed. Confidence: medium.

## Concern 3 — Automated Traffic Pattern

The 575 outbound packets on the 10.10.0.33:53480 → 18.184.26.60:80 flow arrived within about five seconds, a pattern that could represent automated software behaviour. Confidence: low-medium.

## Confirmed Observations

The following are confirmed from the dataset:

- Internal host identified: 10.10.0.33
- Main external communication: 18.184.26.60
- Primary protocol: TCP
- Primary service: HTTP port 80
- Significant HTTPS communication: 204.79.197.200:443
- DNS communication: 8.8.8.8:53
- Total captured traffic: 896 KB

## Potential Security Concerns and Confidence Level

| Finding                                     | Confidence | Reason                                  |
|---------------------------------------------|------------|-----------------------------------------|
| Heavy communication with single external IP | Medium     | Dominant external relationship          |
| Possible automated application traffic      | Low-Medium | High repetition but short duration      |
| HTTP data transmission                      | Medium     | Unencrypted protocol                    |
| Possible C2-like behaviour                  | Low        | No periodic beacon pattern confirmed    |
| Data exfiltration                           | Low        | No evidence of abnormal outbound volume |
| DNS abuse                                   | Low        | Minimal DNS activity                    |

## Overall Security Assessment

Traffic from 10.10.0.33 toward external systems looks normal on its face but is heavily concentrated — the strongest single observation is the very large relationship with 18.184.26.60 over TCP/80 HTTP. That's worth validating either way, since a single external endpoint handling most of a host's traffic can just as easily be a legitimate application or web service connection as unauthorized automated communication. Based on this dataset alone, though, there's no confirmed malware activity, no confirmed C2 beaconing, and no confirmed data exfiltration.

## Recommended Next SOC Investigation Steps

1. Investigate external IP reputation — check VirusTotal, AbuseIPDB, WHOIS, and ASN ownership for 18.184.26.60.
2. Analyze HTTP contents — review the PCAP payloads for URLs, headers, User-Agent strings, POST requests, and any uploaded files.
3. Identify the internal host — confirm hostname, running processes, installed applications, recent downloads, browser history, and scheduled tasks for 10.10.0.33.
4. Perform endpoint investigation — look for unknown executables, persistence mechanisms, suspicious processes, and antivirus alerts.
5. Compare against baseline — determine whether 10.10.0.33 → 18.184.26.60 is expected business or application traffic.

## Final Analyst Conclusion

**Severity assessment:** Low to Medium concern

The dataset mainly shows concentrated web communication from one internal host. It's worth investigating because of the dominant external connection and heavy HTTP usage, but the available evidence isn't enough to call the activity malicious. Packet payload inspection, IP reputation checks, and endpoint analysis are still needed before making that call.

## 5. Analyst Review of AI Output

The AI-generated findings were checked against the dataset directly.

### Correct AI Observations

#### Internal IP classification

It classified 10.10.0.33 as internal using the standard RFC 1918 private ranges — 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

#### External communication identification

It correctly identified 18.184.26.60, 204.79.197.200, 93.184.220.29, and 8.8.8.8 as external/public addresses.

#### Protocol identification

It also correctly identified TCP, HTTP, HTTPS, DNS, and NBNS in the traffic.

## 6. Limitations of AI Analysis

The AI's analysis was based on packet metadata only, without payload inspection, malware signature analysis, endpoint information, or threat intelligence verification. Its traffic patterns should be treated as indicators, not confirmed threats.



## 7. Segment 2 Summary

| Category              | Observation                        |
|-----------------------|------------------------------------|
| Primary internal host | 10.10.0.33                         |
| Main external IP      | 18.184.26.60                       |
| Main protocol         | TCP                                |
| Main service          | HTTP Port 80                       |
| Repeated pattern      | High-volume communication          |
| Beaconing             | Not confirmed                      |
| Risk level            | Low-Medium, investigation required |

## 8. Conclusion

ChatGPT picked out the major communication patterns in Segment 2 — the concentrated traffic between 10.10.0.33 and 18.184.26.60, the heavy HTTP usage, and the repeated packet exchanges. The available metadata wasn't enough to confirm malicious activity on its own; that would still need packet payload inspection, threat intelligence, and endpoint investigation.